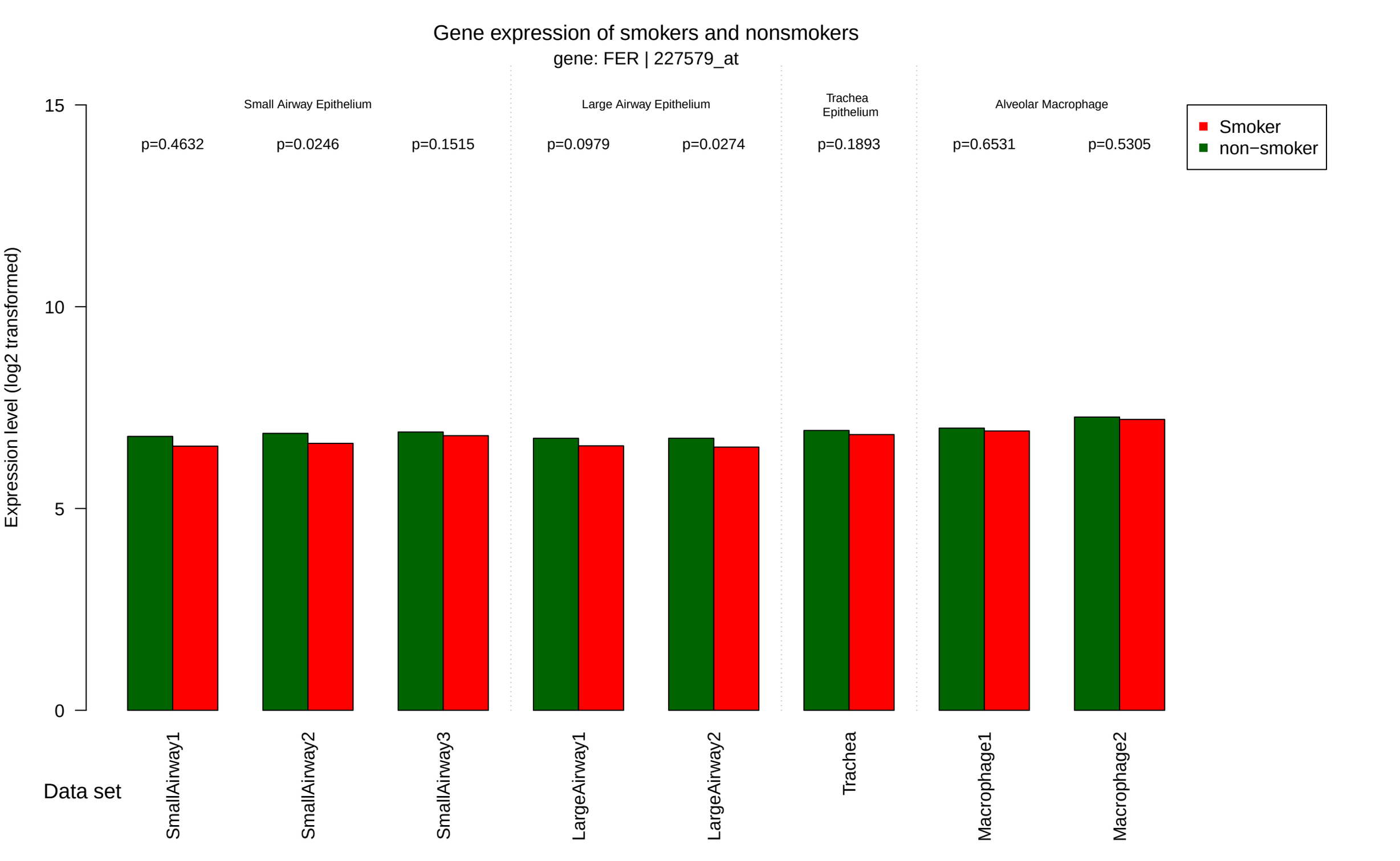
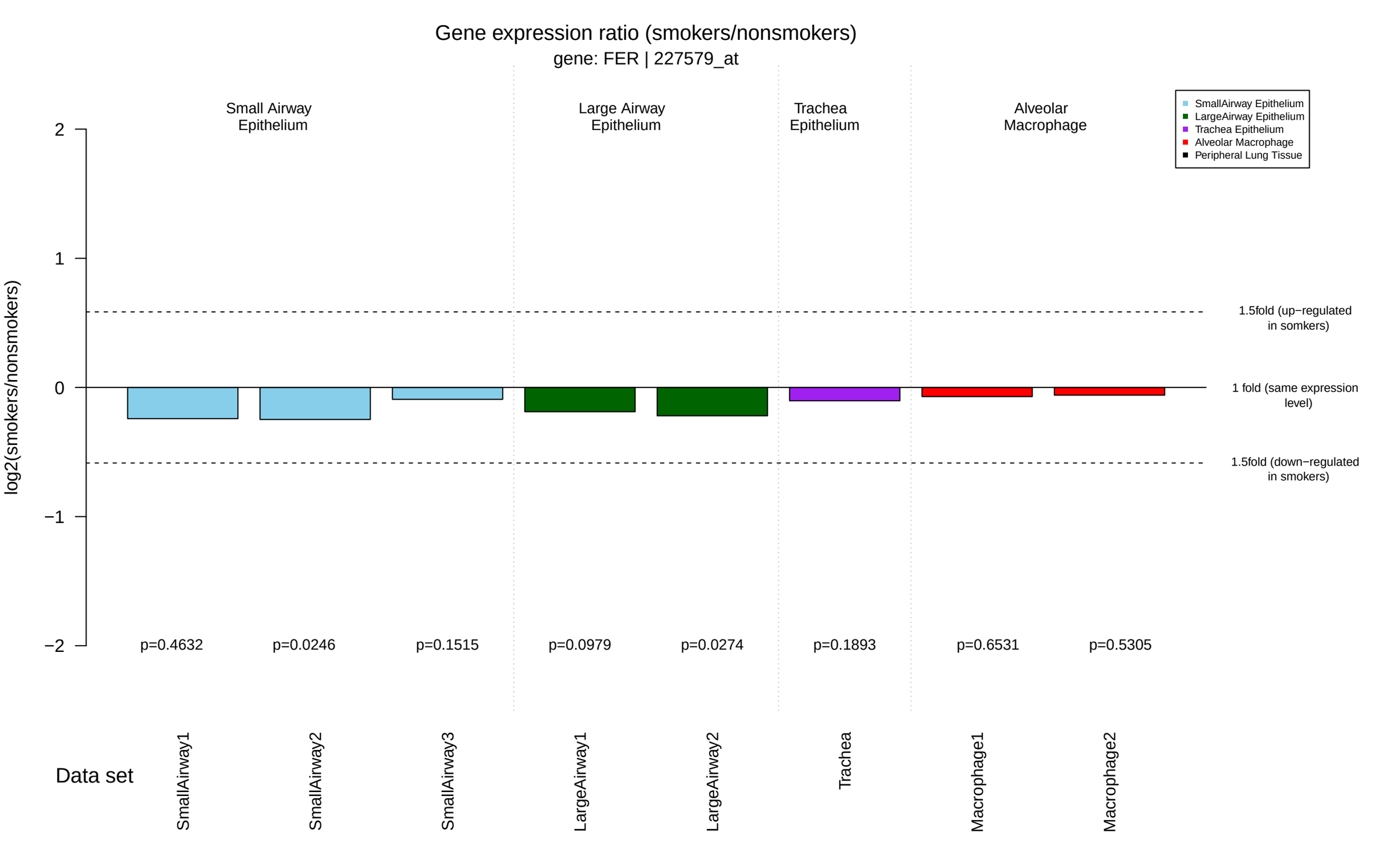
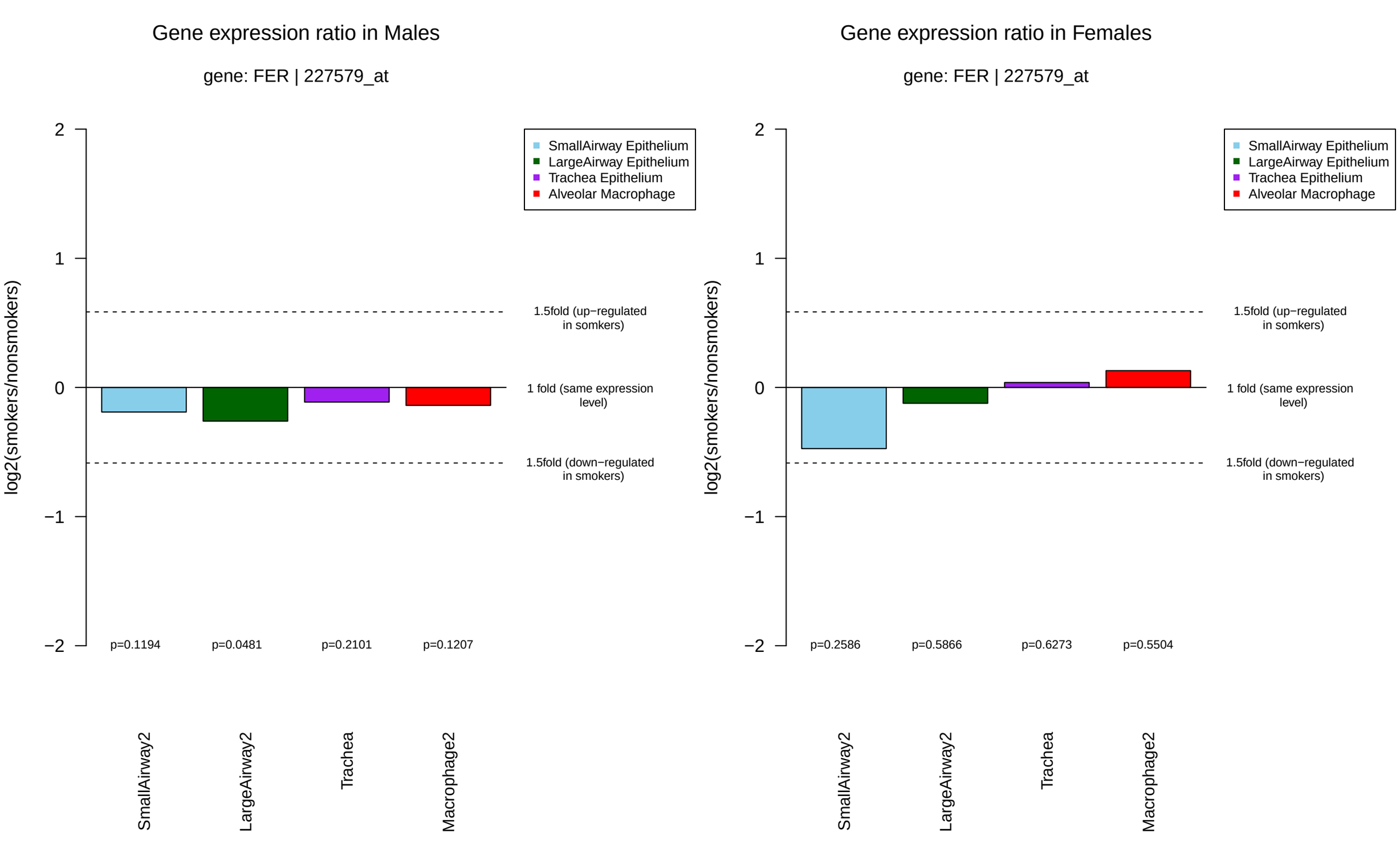
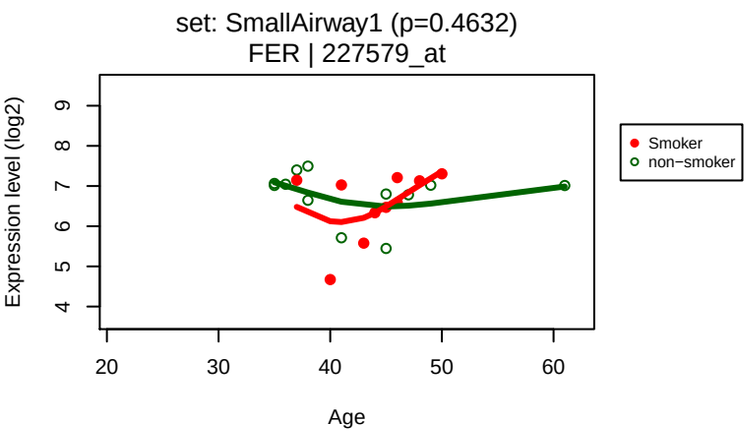
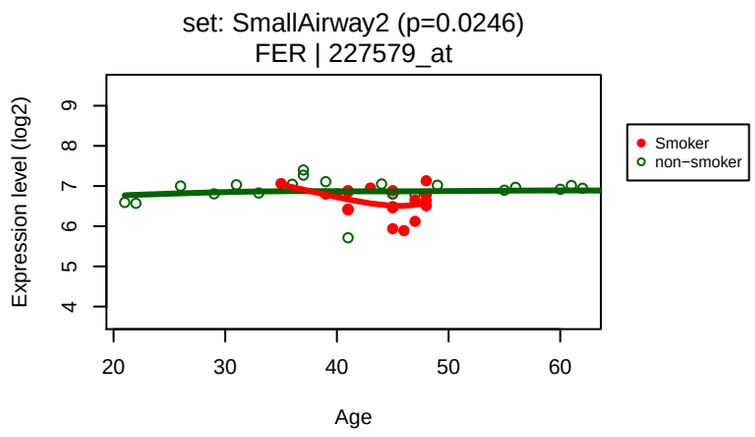
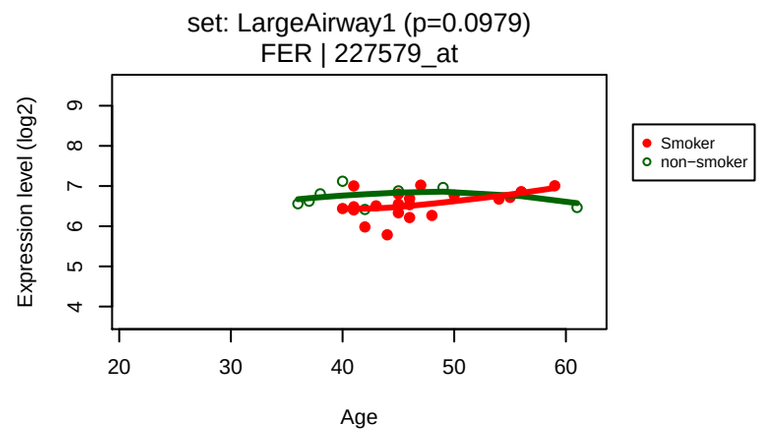

Gene expression of smokers and nonsmokers
gene: FER | 227579_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4632
p=0.0246
p=0.1515
p=0.0979
p=0.0274
p=0.1893
p=0.6531
p=0.5305
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

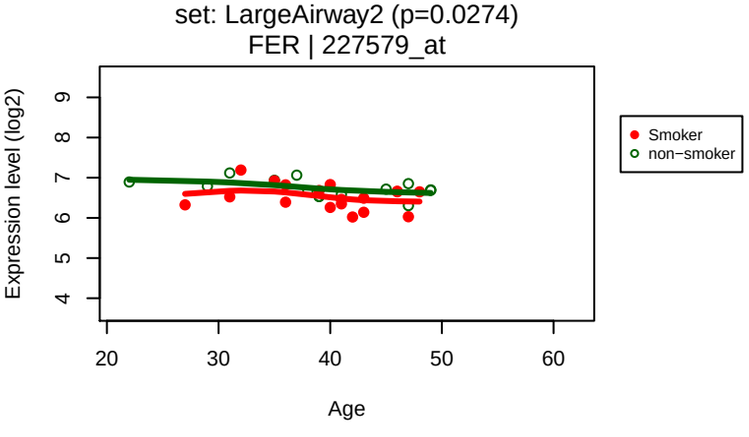

Gene expression ratio (smokers/nonsmokers)
gene: FER | 227579_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4632
p=0.0246
p=0.1515
p=0.0979
p=0.0274
p=0.1893
p=0.6531
p=0.5305
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

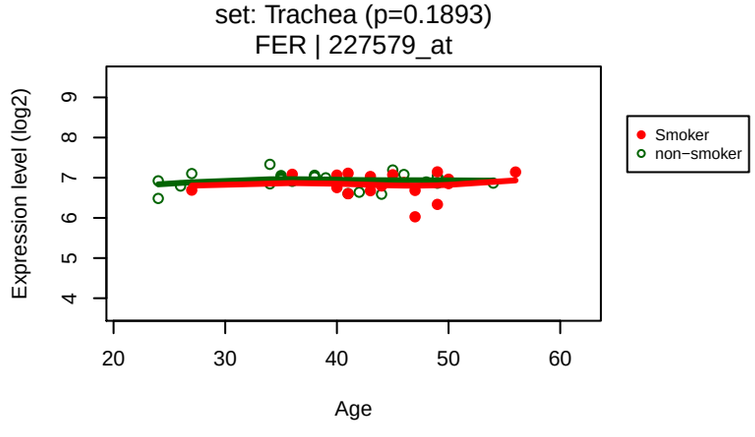

Gene expression ratio in Males
gene: FER | 227579_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1194
p=0.0481
p=0.2101
p=0.1207
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FER | 227579_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2586
p=0.5866
p=0.6273
p=0.5504
SmallAirway2
LargeAirway2
Trachea
Macrophage2

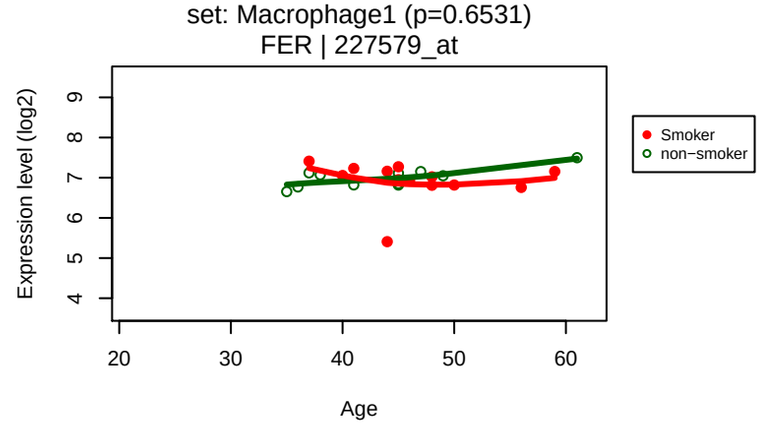

set: SmallAirway1 (p=0.4632)
FER | 227579_at
Expression level (log2)
Age
Smoker
non−smoker

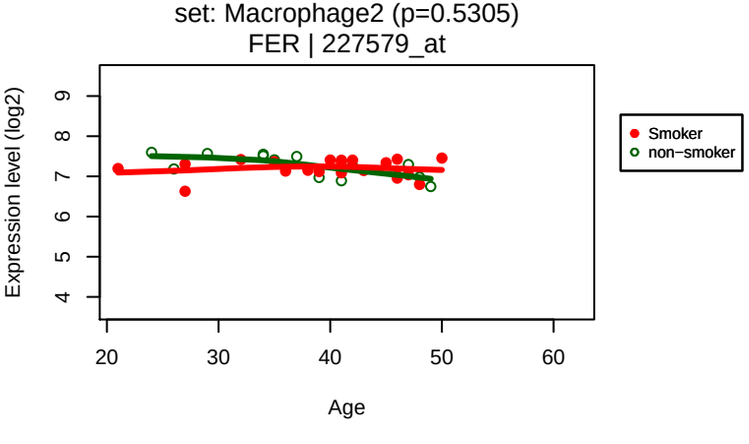

set: SmallAirway2 (p=0.0246)
FER | 227579_at
Expression level (log2)
Age
Smoker
non−smoker

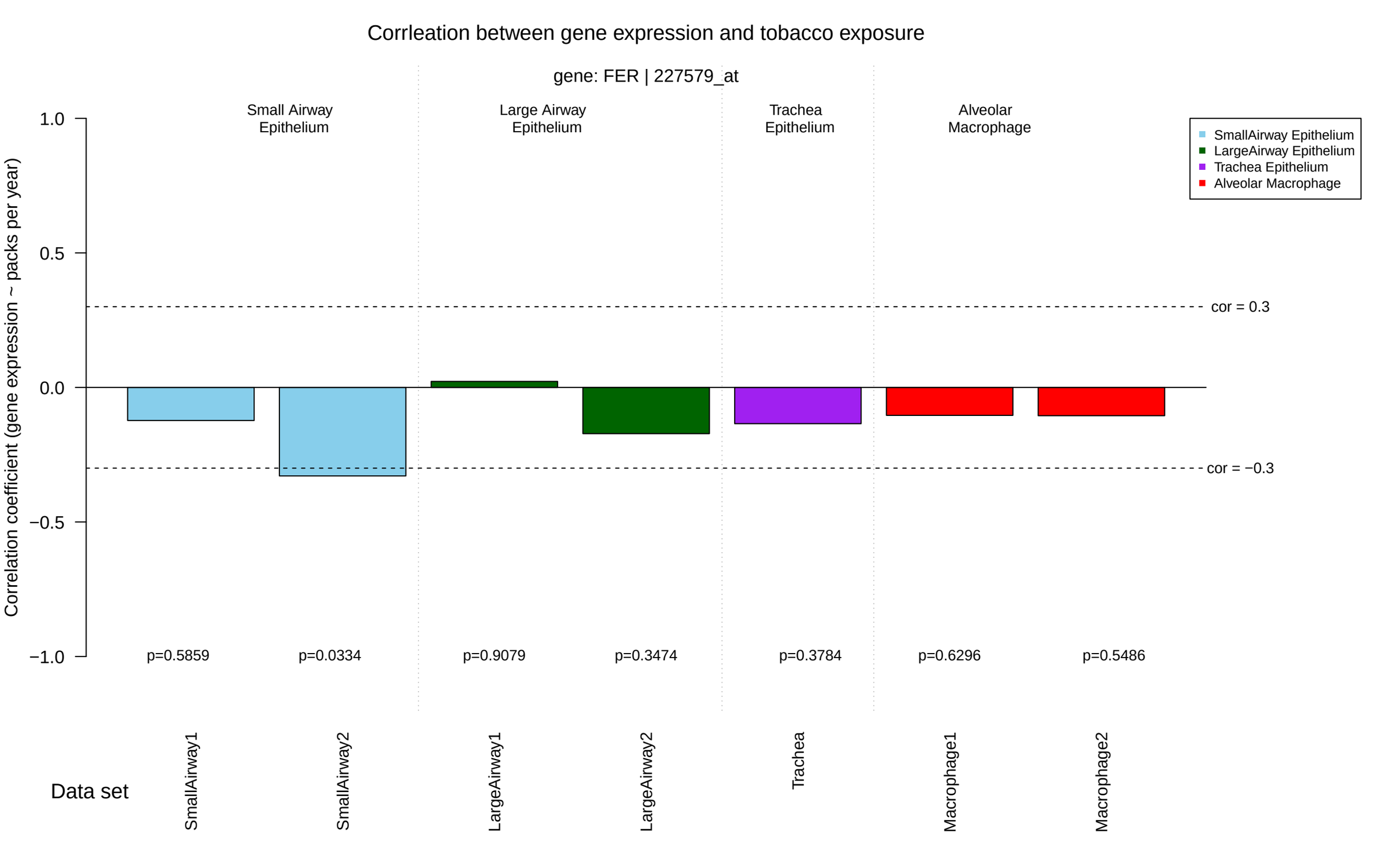

set: LargeAirway1 (p=0.0979)
FER | 227579_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0274)
FER | 227579_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1893)
FER | 227579_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6531)
FER | 227579_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5305)
FER | 227579_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FER | 227579_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5859
p=0.0334
p=0.9079
p=0.3474
p=0.3784
p=0.6296
p=0.5486
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2